HARSH GAJRA
Senior Software Engineer
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Over 10 years as a Staff/Senior Software Engineer, I specialize in building data-intensive, distributed applications across the full SDLC. I lead and mentor teams, consistently recognized for technical and soft skills, enabling effective cross-functional coordination and timely, high-quality project delivery. Seeking a Technology Lead or Engineering Manager role to leverage both technical depth and leadership skills to build resilient systems and drive business impact.

WORK EXPERIENCE

Senior Software Engineer – Crux Data (Feb 2023 – Present)

- Designed and implemented distributed, event-driven, scalable APIs for Crux's Data Platform using FastAPI, Pub/Sub, and PostgreSQL. These were containerized in Docker, orchestrated via Kubernetes, and deployed on GCP for high availability and reliability.
- Migrated over 2 billion Cassandra records to AlloyDB (PostgreSQL) using Python, Bash, SQL, Kubernetes, BigQuery and GCS, cutting annual infrastructure costs by over $400K.
- Architected Git-driven CI/CD framework automating complex data workflows via YAML, GitHub Actions, and merge queues, reducing pipeline deployment downtime by 1,800 hours annually.
- Led a team of 5 software engineers, providing technical guidance and mentorship to ensure successful project execution and delivery.
- Lead Engineer for cruxctl, a Python CLI tool built with the Typer framework, providing customers with command-line access to Crux's External data platform.
- Accelerated onboarding and reduced manual tuning of Apache Airflow ingestion pipelines by developing an AI-powered adapter recommendation engine using OpenAI GPT LLMs to automate configuration.
- Mentored data engineers as SME in designing complex, version-controlled ETL manifests integrated via Git-based CI/CD and deployed with Apache Airflow, contributing to the core functionality of Crux's ETL service.
- As part of Platform Engineering, built and deployed asynchronous Java microservices with Spring Boot, PostgreSQL, Docker and Google Cloud Platform.

Software Engineer - Amazon Web Services, Data Migration Service (Sept,2021 – Feb 2023)

- Developed a distributed serverless test execution platform using AWS Lambda, AWS SWF, AWS CloudWatch, and CDK with Java 11. This platform scales to manage and execute hundreds of thousands of parallel test cases for AWS DMS components (Control Plane, Data Plane, Serverless).
- Re-architected Canary and Test infrastructure, resulting in the elimination of deployment bottlenecks and congestion, leading to overall stabilization, faster deployments, and an estimated savings of 50 hours per week.
- Started as 1st hire on the Platform STP team, mentored and led a team of 3 interns and 2 engineers, and served as subject matter expert for backend components of the DMS service.
- Built on-call support automation for the AWS DMS product which reduced on-call tickets by 60% weekly and saved over 2,400 hours annually.

Software Developer 2 – FINRA (May 2020 - Sept, 2021)

- Designed and developed the core components of FINRA's Consolidated Audit Cat (CAT) platform, enabling the capture and traceability of billions of transactions across multiple U.S. stock exchanges to meet SEC regulatory mandates.
- Developed and optimized advanced SQL queries to process and analyze high-volume market data, producing actionable reports and insights that enhanced project dashboards used by leadership for operational decision-making.

Software Engineer - Pype Inc (Acquired by AutoDesk in May 2020) (July 2018 – May 2020)

- Architected and developed the backend for Pype's Autospecs and eBinder software products using Java 8, Spring Boot, JPA, Redis,RabbitMQ and MySQL databases.
- Designed and developed software solutions for optimizing the business process followed by in-house sales and support teams thus saving the organization $360,000 per year in costs and 15,000 man-hours per year.
- Developed, analyzed, and implemented algorithms in Java for extracting submittal information from construction drawing sheets.

TECHNICAL SKILLS
Programming Skills: Python 3, Java, Bash, FastAPI, Pub/Sub, Kafka, Typer, AWS, Spring Boot.
Databases: Amazon RDS, Postgres, AlloyDB, Cassandra, Amazon Aurora, MySQL, DynamoDB, BigQuery.
Tools: GitHub, JIRA, Apache Airflow, GPT, LLMs, AI, GitHub Actions, Docker, Kubernetes, Co-Pilot, Cursor.
Other: Brazil Build System, AWS CDK, SWF, AWS, Cloudwatch, EC2, GCP, CodeFresh CI/CD.

EDUCATION
Masters in Computer Science - State University Of New York, Binghamton, August 2016 - May 2018
Bachelor of Computer Engineering - K.J.Somaiya College of Engineering, Mumbai, First class (2011- 2015)

PAPER PUBLICATIONS
Paper: Credit card Fraud Detection and Prevention Using Perceptron Training Algorithm.
Journal: International Journal on Recent and Innovation Trends in Computing and Communication, Apr 2015 Link: https://ijritcc.org/index.php/ijritcc/article/view/4182/4182